Saarthi Manual

Section 8: Real-World Testing, Calibration & Final Adjustments

🎯 8.1 OBJECTIVE

By the end of this section, you will:
✅ Conduct safe and meaningful field tests
✅ Identify and fix real-world performance issues
✅ Calibrate sensor angles and thresholds
✅ Validate stair-down detection
✅ Prepare test data to include in your project report/demo

🚶 8.2 PREPARING FOR FIELD TESTING

Before testing, verify:

Test Readiness	Status
All components working	✅
Device waterproofed	✅
Battery fully charged	✅
Speaker loud & clear	✅
Voice alerts working	✅

⚠️ **Carry a backup power bank and keyboard+mouse (if no SSH)**

🧪 8.3 TEST SCENARIOS TO PERFORM

🧱 A. Stationary Obstacle (Front Detection)

Test	Action
1.	Place a chair ~1m ahead. Observe camera alert.
2.	Walk slowly toward chair. Ensure "Obstacle very close" plays near 50cm.

✅ *Expected Output*:

- “Chair ahead” or “Object ahead”
- “Obstacle very close” at <50cm

🐾 B. Moving Objects (Dynamic Scene)

Test	Action
1.	Have a person or dog walk into view of the camera.
2.	Ensure dynamic object is detected without stutter.

✅ *Expected Output*:

- "Person ahead" or "Dog ahead"

⬇️ C. Stair Down / Pit Detection

Test	Action
1.	Hold cane over stairs, pit, or edge.
2.	Ensure the **downward sensor** triggers only when there's no reflected surface.

✅ *Expected Output*:

- "Watch out! Stairs going down"
- No false alarms on flat ground

✅ *Sensor Calibration*:

- If downward detection happens too early → **reduce trigger height**
- If it never triggers → **lower the sensor angle** to ~45–60°

🌧️ D. Rainy Conditions or Water Splash Test

| Test | Simulate light water on stick (do **not** soak the pouch). |
| ✅ *Expected Output*: |

- Saarthi should function without false alerts
- No water damage or shorting

✅ *If you're afraid to use water*: Use a **spray bottle** to test waterproofing in a controlled way

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🔍 8.4 TWEAKING SENSOR THRESHOLDS

In `main.py`:

```
# Adjust as needed based on field tests
if front_distance < 50:
    speak("Obstacle very close")

if down_distance > 100:
    speak("Watch out! Stairs going down")
```

Issue	Fix
False “stairs down” alerts	Lower angle or reduce threshold to 80 cm
Doesn't detect stair edges	Raise cane higher OR increase threshold
Too many object alerts	Add `time.sleep()` cooldown or filter labels

📈 8.5 DOCUMENTING TEST RESULTS

Make a test report table like this (for submission/demo):

Test Type	Location	Input	Expected Output	Actual Output	Notes
Front Object	Home Hallway	Chair at 1m	“Chair ahead”	Works ✅	Good
Pit/Stairs	School steps	Drop after floor	“Stairs going down”	Works ✅	Great
Rain	Balcony rain	Light drizzle	No failure, continues alert	Works ✅	Safe

✅ Save 4–5 such logs and photos/videos for your **final demo or presentation**

🧼 8.6 BONUS: FINAL POLISH

✅ 1. Remove Unnecessary Voice

Modify `main.py` to only speak important things.

```
if obj in ["person", "dog", "car"]:  # Avoid saying "bottle", "banana"
    speak(f"{obj} ahead")
```

✅ 2. Add Logs (for debugging later)

```
with open("log.txt", "a") as log:
    log.write(f"{time.ctime()} - {obj}\n")
```

✅ 3. Battery Monitoring (Advanced)

You can use a battery with voltage indicator or add a USB power sensor inline.

✅ 8.7 SECTION SUMMARY

You now have:

- Tested your cane in **real-life scenarios**
- Tuned sensor angles and thresholds
- Verified **stair detection + waterproofing** — beating WeWALK
- Created a **report-ready demo system**

🏁 WHAT'S NEXT?

🎓 **Final Section (Section 9)**:
Documentation, Demo Preparation & Submission-Ready Packaging

In it, we'll create:

- Abstract
- User manual
- Demo script
- Wiring diagrams
- Final GitHub + PDF files